

NETCAT (NC) CHEAT SHEET

The Swiss Army Knife of Networking & Security Testing

One-Line Summary: Ncat is a flexible networking utility used to create TCP/UDP connections, listen on ports, transfer files, and perform basic network scanning & diagnostic testing.

01. CORE ARCHITECTURE & MENTAL MODEL

Netcat connects two endpoints and enables bi-directional data flow between them. It operates at the network layer and can act as either a client or a listener/server.

Client Mode (Active)

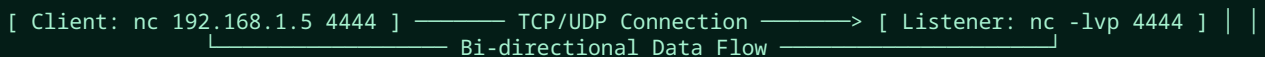
```
nc <IP> <PORT>
```

- Initiates an outbound connection to a target remote host and listening port.

Listener Mode (Passive)

```
nc -lvp <PORT>
```

Binds to a local port and waits for incoming connections from remote clients.



02. QUICK COMMAND REFERENCE

COMMAND	PURPOSE / DESCRIPTION
<code>nc IP PORT</code>	Connect to a specific target IP address and port.
<code>nc -lvp PORT</code>	Listen for incoming connections (Listen, Verbose, Port).
<code>nc -zv IP PORT</code>	Check if a specific port is open without sending normal data.
<code>nc -zv IP PORT_RANGE</code>	Scan a range of ports (e.g., <code>20-100</code>).
<code>nc IP PORT < file</code>	Send file content from client to listener.
<code>nc -lvp PORT > file</code>	Receive incoming stream and save directly to file.
<code>nc -u IP PORT</code>	Establish connection using UDP instead of default TCP.

03. ESSENTIAL FLAGS & OPTIONS

-1 (Listen)

Tells Netcat to wait for an incoming connection rather than initiating one.

-v (Verbose)

Provides detailed output regarding connection attempts and status updates.

-p (Port)

Specifies the local port number to bind to in listener mode.

-z (Zero-I/O)

Sets scan mode: checks port status without sending interactive session data.

04. PRIMARY USE-CASE WORKFLOWS

A. File Transfer

1. Receiver Starts Listener First:

```
nc -lvp 4444 > received.txt
```

2. Sender Transmits File:

```
nc 192.168.1.5 4444 < file.txt
```

 **Security Note:** Data transferred via Netcat is completely unencrypted. Avoid sending sensitive information over public networks.

B. Port Scanning & Enumeration

Check single port status:

```
nc -zv 192.168.1.10 80
```

Scan a defined port range:

```
nc -zv 192.168.1.10 20-100
```

 Useful for quick checks, though dedicated tools like Nmap offer comprehensive scanning capabilities.

C. Protocol Interaction (Banner Grabbing)

Manually talk to network protocols (HTTP, SMTP, FTP):

```
nc 192.168.1.10 80
GET / HTTP/1.1
Host: example.com
```

Common Protocol Ports: **FTP: 21**

SSH: 22 **SMTP: 25** **HTTP: 80**

D. Reverse Shell Concept (Lab / Testing)

Target initiates a connection back to listener:

```
Target Machine — Outbound Connection —>
Analyst Listener (Initiates shell) (nc -lvp 4444)
```

* Strictly for use in authorized lab environments and penetration tests.

05. NETCAT VS. TELNET

FEATURE	TELNET	NETCAT (NC)
Primary Role	Remote terminal protocol / client	General-purpose networking utility
Default Port	TCP Port 23	Flexible (connects to / listens on any port)
Capabilities	Text-based remote administration	Listener, Client, File Transfer, Port Scanning, UDP Support

06. ★ COMMANDS TO REMEMBER

```
nc IP PORT                # Connect to a target port
nc -lvp PORT               # Start listening on a local port
nc -zv IP PORT             # Check if a single port is open
nc -zv IP 20-100           # Scan range of ports
nc IP PORT < file.txt      # Transfer file to listener
nc -lvp PORT > file.txt    # Receive file stream and save
nc -u IP PORT              # Establish UDP session
```